3. Method

Bookmark this page

Consider a first-order homogeneous linear system of ODEs:

$$\dot{\mathbf{x}} = \mathbf{A}\mathbf{x},$$

where $\mathbf{A}$ is an $n \times n$ matrix with constant, real entries.

Recall (from the course *Differential equations: 2 by 2 systems*) that $\mathbf{v}e^{\lambda t}$ is a solution if and only if $\mathbf{v}$ is an eigenvector of $\mathbf{A}$ with eigenvalue λ.

Reason

If $\mathbf{x} = \mathbf{v}e^{\lambda t}$, then $\dot{\mathbf{x}} = \lambda\mathbf{v}e^{\lambda t}$ and $\mathbf{A}\mathbf{x} = \mathbf{A}\mathbf{v}e^{\lambda t}$. Therefore,

$$\begin{aligned} & \dot{\mathbf{x}} = \mathbf{A}\mathbf{x} \\ \Longleftrightarrow\ & \lambda\mathbf{v}e^{\lambda t} = \mathbf{A}\mathbf{v}e^{\lambda t} \qquad (\text{for all } t). \\ \Longleftrightarrow\ & \lambda\mathbf{v} = \mathbf{A}\mathbf{v} \qquad (\text{cancel } e^{\lambda t} \text{ from both sides}). \end{aligned}$$

Note that we are following the convention that in expressions like $\lambda\mathbf{v}e^{\lambda t}$, scalar functions such as $e^{\lambda t}$ are placed to the right, while constant scalars and constant vectors are placed to the left.

Conclusion: $\mathbf{v}e^{\lambda t}$ is a solution if and only if $\mathbf{v}$ is an eigenvector of $\mathbf{A}$ with eigenvalue λ.

Hide

Steps to find a basis of solutions to $\dot{\mathbf{x}} = \mathbf{A}\mathbf{x}$, given an $n \times n$ matrix $\mathbf{A}$:

1. Find the eigenvalues of $\mathbf{A}$. These are the roots of the characteristic polynomial $\det(\lambda\mathbf{I} - \mathbf{A})$.

2. For each eigenvalue λ:

- Find a basis for the corresponding eigenspace $\mathrm{NS}(\lambda\mathbf{I} - \mathbf{A})$. Call these basis vectors $\mathbf{v}_1, \cdots, \mathbf{v}_k$.
- Each vector-valued function $\mathbf{v}_i e^{\lambda t}$ is a solution. A solution of this form is called a **normal mode.**

3. If n such solutions were found (i.e., the sum of the dimensions of the eigenspaces is n), then these n solutions are enough to form a basis of all solutions.

Remark 3.1 The solutions of this type will automatically be linearly independent, since their values at $t = 0$ are linearly independent. (The chosen eigenvectors within each eigenspace are independent, and there is no linear dependence between eigenvectors with different eigenvalues.)

Remark 3.2 Note that λ and $\mathbf{v}$ may be complex, which means that eventually you will want to find a basis of real solutions.

Remark 3.3 The only thing that could go wrong is this: if there is a repeated eigenvalue λ, and the dimension of the eigenspace of λ is less than the multiplicity of λ, then the method above does not produce enough solutions. We will not deal with this case until the next lecture.

3. Method

Hide Discussion

Topic: Unit 3: Solving systems of first order ODEs using matrix methods / 3. Method

Add a Post

Show all posts by recent activity

There are no posts in this topic yet.

×

Previous Next